

CDS1114

Introduction to Digital Systems

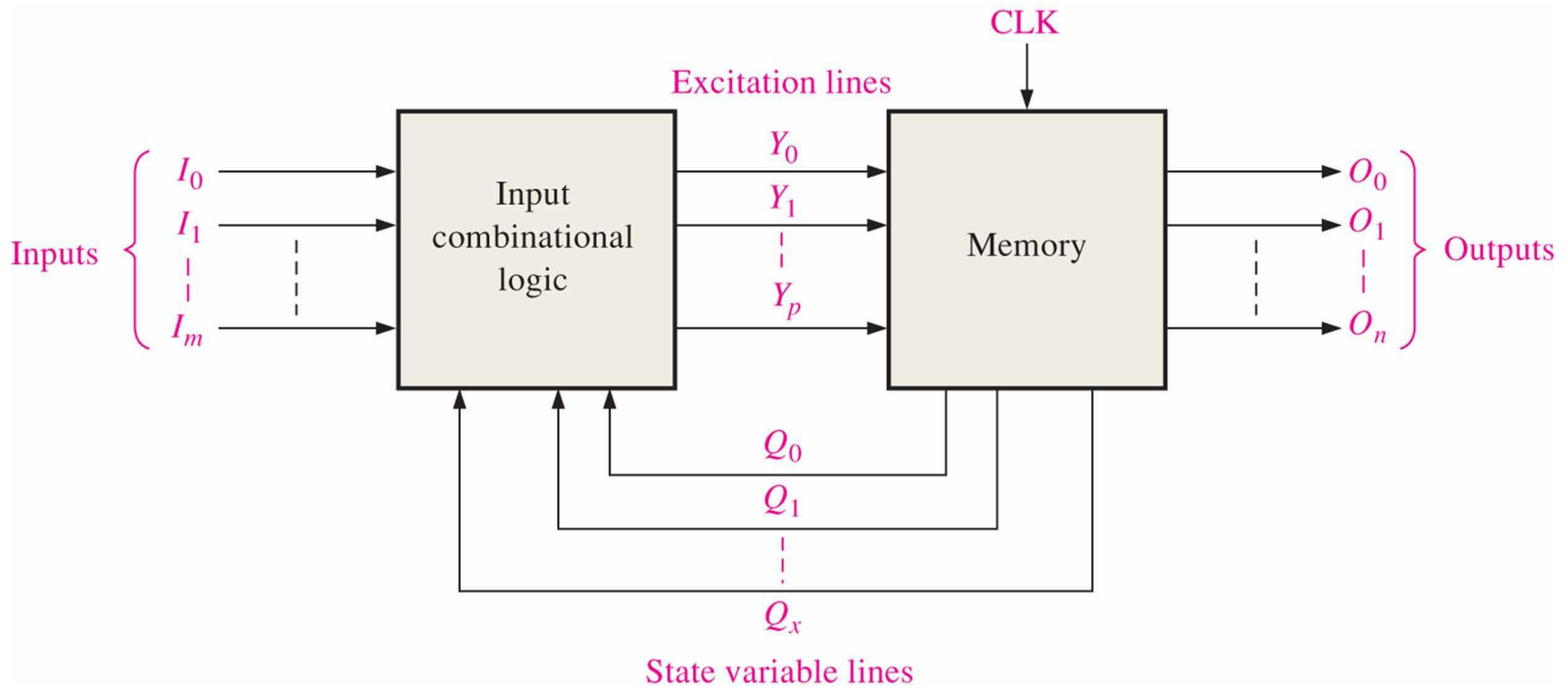
Counters II

Lecture outcome

- ∞ By the end of today's lecture you should be able to
 - design your own synchronous counter that follows a specific pattern
 - implement the design as a logic circuit
- ∞ **NOTE:** contents in this set of slides are intentionally incomplete and content will be shown in class as examples of how they are derived

Synchronous Counter Design

- ✧ Most requirements for synchronous counters can be met with available ICs.
- ✧ In cases where a special sequence is needed, you can apply a step-by-step design process
- ✧ Basically, you start with the desired sequence and draw a state diagram and next-state table, simplify the states using algebra rules and/or k-map and complete the logic circuit
 - The resulting circuit will be a sequential circuit or state machine
- ✧ In general, a sequential circuit consists of a combinational logic section and a memory section (flip-flop)
- ✧ In the case of a clocked sequential circuit, a CLK input to the memory is added



Synchronous Counter Design

Several methods are available that follow arbitrary sequence.

Here we will learn one common method using JK flip-Flops.

In synchronous counters all the FF's are clocked at the same time.

J-K Excitation Table

Before begin the designing we must know the operation of the J-K FF, let us analysis **Truth table for 74LS76 IC (JK flip-flop) and its excitation table.**

Truth table for 74LS76 IC (JK flip-flop)

$\overline{CLEAR}$	$\overline{PRESET}$	J	K	CLK	Qt	Qt'
0	0	0	0	x	1	1
		0	1	x	1	1
		1	0	x	1	1
		1	1	x	1	1
0	1	0	0	x	1	0
		0	1	x	1	0
		1	0	x	1	0
		1	1	x	1	0
1	0	0	0	x	0	1
		0	1	x	0	1
		1	0	x	0	1
		1	1	x	0	1
1	1	0	0	↓	Q ₀	Q ₀ '
		0	1	↓	0	1
		1	0	↓	1	0
		1	1	↓	TOGGLE	TOGGLE

JK FF Excitation Table where $\overline{CLEAR}$ and $\overline{PRESET}$ are High (1)

PRESENT	NEXT	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

OUTPUT TRANSITIONS		FLIP-FLOP INPUTS	
Q_N	Q_{N+1}	J	K
0	→ 0	0	X
0	→ 1	1	X
1	→ 0	X	1
1	→ 1	X	0

J-K Excitation Table

TRANSITION AT OUTPUT		PRESENT STATE Q(N)	NEXT STATE Q(N+1)	J	K
0	0	0	0	0	X
0	1	0	1	1	X
1	0	1	0	X	1
1	1	1	1	X	0

0 → 0 TRANSITION; FF's Present status is 0 and it should remain in 0 when a clock pulse is applied. That can be either J=K=0 status or J=0,K=1.

That means J=0 and K=0 or 1

Resulted , J=0 and K=X(don't care)

Synchronous Counter Design

0 → 1 TRANSITION: The present state is 0 and it has to change to 1. This can happen either $J=1$ and $K=0$ or $J=K=1$.

That means $J=1$ and $K=0$ or 1

Resulted , $J=1$ and $K=X$ (don't care)

1 → 0 TRANSITION; The present state is 1 and it has to change to 0. This can happen either $J=0$ and $K=1$ or $J=K=1$.

That means $K=1$ and $J=0$ or 1

Resulted , $K=1$ and $J=X$ (don't care)

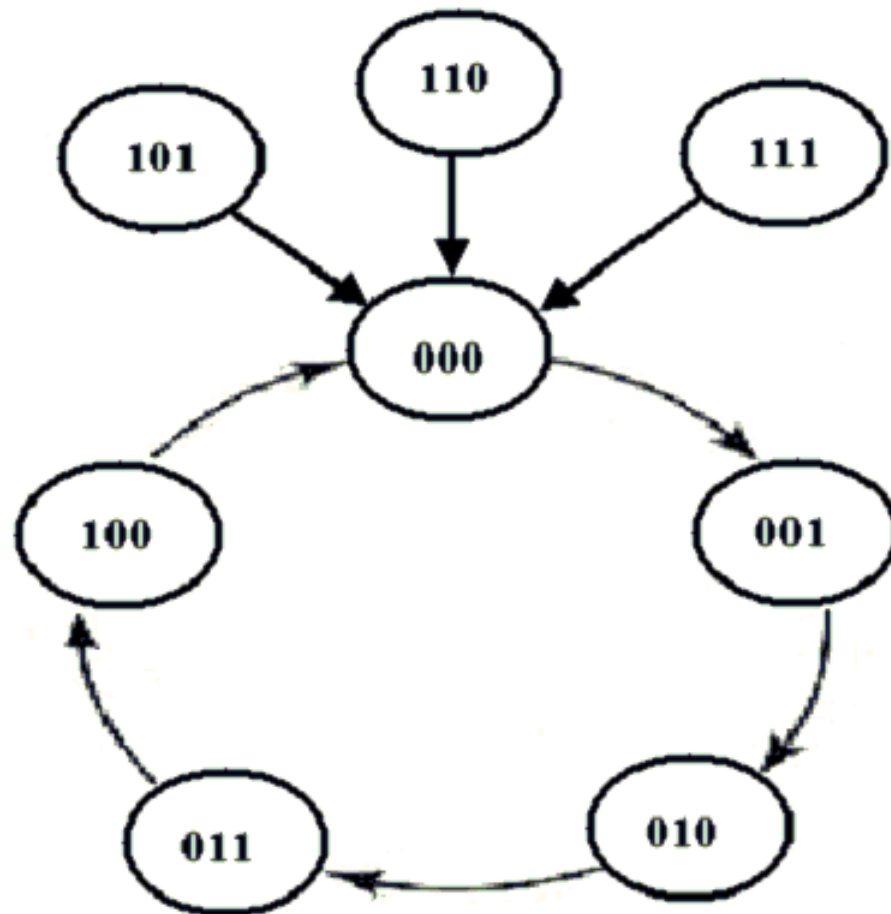
1 → 1 TRANSITION; The present state is 1 and it has to change to 1. This can happen either $J=K=0$ or $J=1$ and $K=0$.

That means $K=0$ and J can be 0 or 1

Resulted , $K=0$ and $J=X$ (don't care)

STEP -1

Draw the state transition diagram showing all the possible states, including those that are not part of the desired counting sequence



Synchronous Counter Design / Example (1)cont.

STEP -2

Use the state transition diagram to set up a table that lists all **PRESENT** states and their **NEXT** states

	Present state			Next state		
	<i>C</i>	<i>B</i>	<i>A</i>	<i>C</i>	<i>B</i>	<i>A</i>
1	0	0	0	0	0	1
2	0	0	1	0	1	0
3	0	1	0	0	1	1
4	0	1	1	1	0	0
5	1	0	0	0	0	0
6	1	0	1	0	0	0
7	1	1	0	0	0	0
8	1	1	1	0	0	0

Undesired states transition to 000 here

STEP -3

Add a column to this table for each J and K input. For each **PRESENT** state, indicate the level required at each J and K input in order to produce the transition to the **NEXT** state.

OUTPUT TRANSITIONS		FLIP-FLOP INPUTS	
Q_N	Q_{N+1}	J	K
0	→ 0	0	X
0	→ 1	1	X
1	→ 0	X	1
1	→ 1	X	0

	Present state			Next state										
	C	B	A	C	B	A	J_C	K_C	J_B	K_B	J_A	K_A		
1	0	0	0	0	0	1	0	X	0	X	1	X		
2	0	0	1	0	1	0	0	X	1	X	X	1		
3	0	1	0	0	1	1	0	X	X	0	1	X		
4	0	1	1	1	0	0	1	X	X	1	X	1		
5	1	0	0	0	0	0	X	1	0	X	0	X		
6	1	0	1	0	0	0	X	1	0	X	X	1		
7	1	1	0	0	0	0	X	1	X	1	0	X		
8	1	1	1	0	0	0	X	1	X	1	X	1		

STEP 4 : K-MAP SIMPLIFICATIONS

- ✎ Determine the logic required for the J and K inputs of each FF in the counter using K-maps
 - There will be one K-map for the J input and another K-map for the K input of each FF
- ✎ From the J and K states in the transition table a 1, 0 or X is mapped onto each present-state cell on the K-maps depending on the transition of the Q output for a particular FF
- ✎ An example of this operation is shown on the following slide

STEP- 4

Design the logic expression to generate the level required at each J and K, using K-maps.

Present state				
<i>C</i>	<i>B</i>	<i>A</i>	J_A	K_A
0	0	0	1	X
0	0	1	X	1
0	1	0	1	X
0	1	1	X	1
1	0	0	0	X
1	0	1	X	1
1	1	0	0	X
1	1	1	X	1

	$\bar{A}$	<i>A</i>
$\bar{B}\bar{C}$	1	X
$\bar{B}C$	0	X
<i>BC</i>	0	X
<i>B\bar{C}</i>	1	X

$$J_A = \bar{C}$$

	$\bar{A}$	<i>A</i>
$\bar{B}\bar{C}$	X	1
$\bar{B}C$	X	1
<i>BC</i>	X	X
<i>B\bar{C}</i>	X	1

$$K_A = 1$$

STEP- 4cont.

Present state				
<i>C</i>	<i>B</i>	<i>A</i>	<i>J_B</i>	<i>K_B</i>
0	0	0	0	X
0	0	1	1	X
0	1	0	X	0
0	1	1	X	1
1	0	0	X	X
1	0	1	X	X
1	1	0	X	1
1	1	1	X	1

	$\bar{A}$	<i>A</i>
$\bar{B}\bar{C}$	0	1
$\bar{B}C$	0	0
<i>BC</i>	X	X
<i>B</i> $\bar{C}$	X	X

$$J_B = A \bar{C}$$

	$\bar{A}$	<i>A</i>
$\bar{B}\bar{C}$	X	X
$\bar{B}C$	X	X
<i>BC</i>	1	1
<i>B</i> $\bar{C}$	0	1

$$K_B = A + C$$

STEP- 4cont

Present state				
<i>C</i>	<i>B</i>	<i>A</i>	<i>J_c</i>	<i>K_c</i>
0	0	0	0	X
0	0	1	0	X
0	1	0	0	X
0	1	1	1	X
1	0	0	0	1
1	0	1	X	1
1	1	0	0	1
1	1	1	X	1

	$\bar{A}$	<i>A</i>
$\bar{B}\bar{C}$	0	0
$\bar{B}C$	X	X
<i>BC</i>	X	X
$B\bar{C}$	0	1

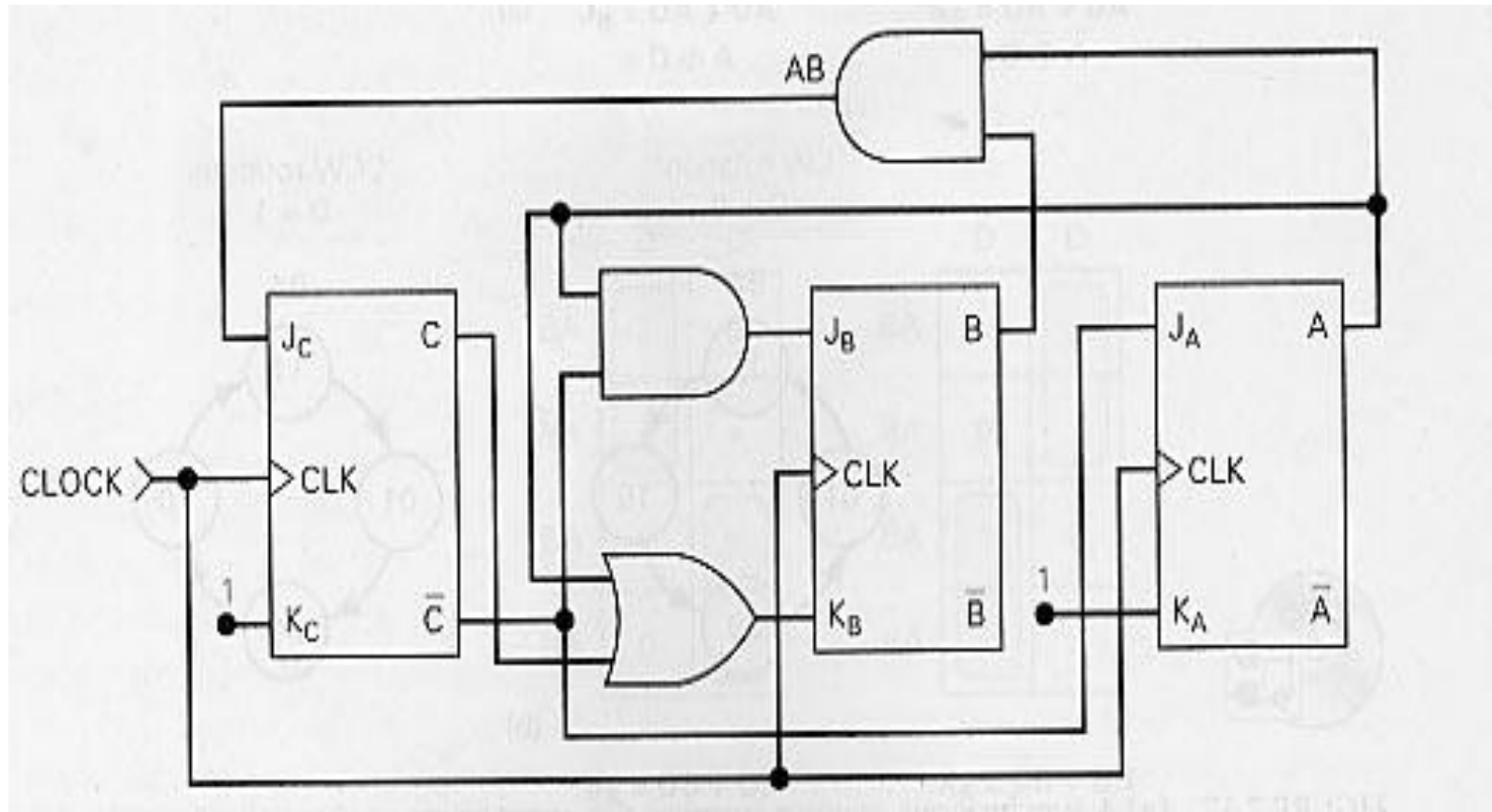
$$J_c = AB$$

	$\bar{A}$	<i>A</i>
$\bar{B}\bar{C}$	X	X
$\bar{B}C$	1	1
<i>BC</i>	1	1
$B\bar{C}$	X	X

$$K_c = 1$$

SETP -5

Finally to implement the final expressions.



$$J_C = AB$$

$$K_C = 1$$

$$J_B = A\bar{C}$$

$$K_B = A + C$$

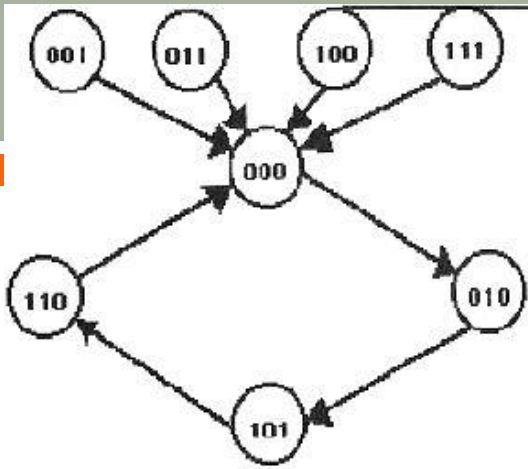
$$J_A = \bar{C}$$

$$K_A = 1$$

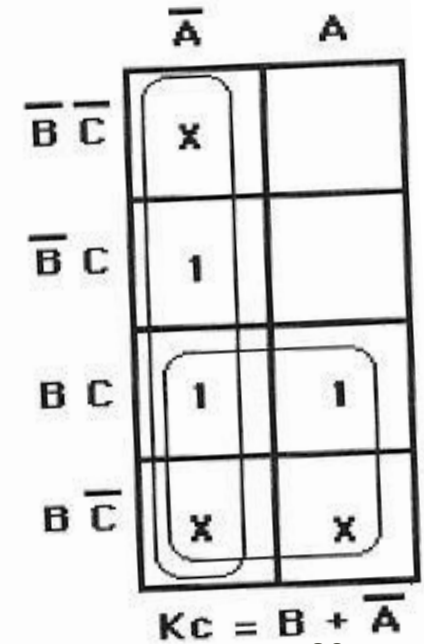
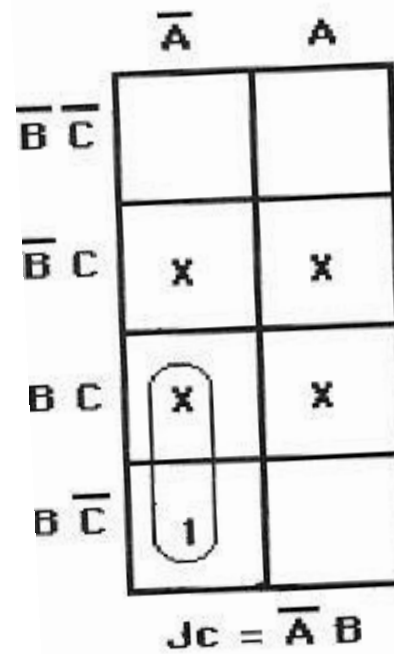
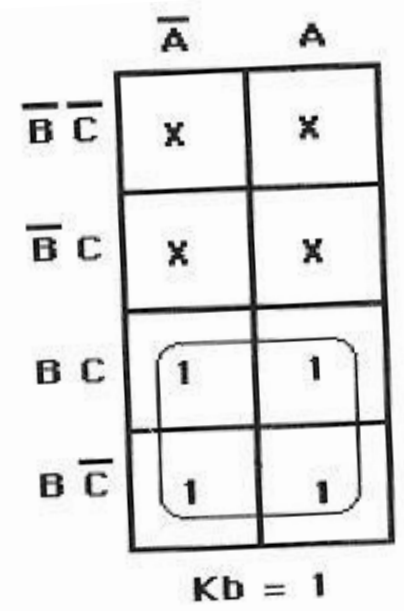
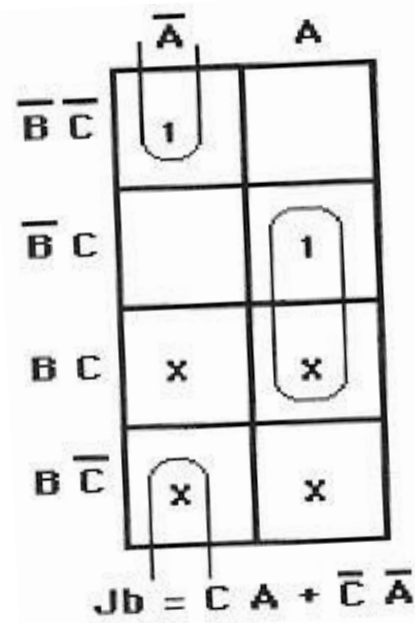
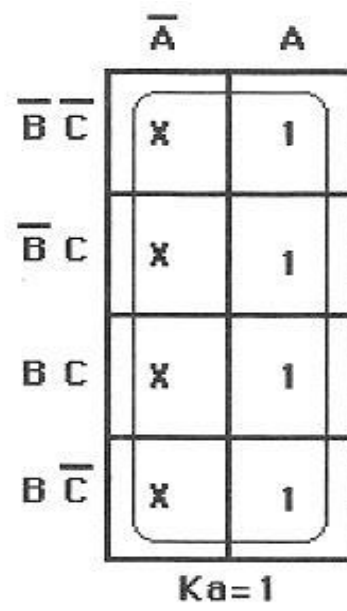
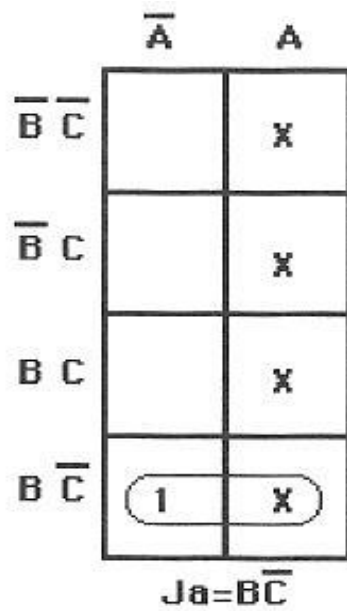
Exercises

1. Design a JK synchronous counter that has follows the sequence 000,010,101,110 and repeats. The undesired states go to 000 on the next clock pulse
2. Design a JK synchronous counter that follows the sequence 000,010,101,110 and repeats. For undesired states their NEXT states can be DON'T CARES

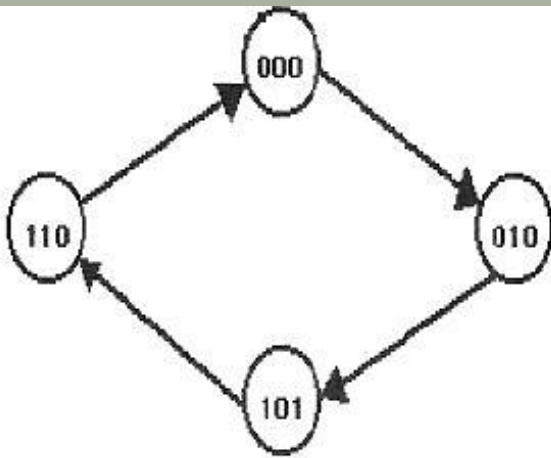
Answer 1



PRESENT STATE			NEXT STATE			FF2		FF1		FF0	
C	B	A	C	B	A	Jc	Kc	Jb	Kb	Ja	Ka
0	0	0	0	1	0	0	X	1	X	0	X
0	0	1	0	0	0	0	X	0	X	X	1
0	1	0	1	0	1	1	X	X	1	1	X
0	1	1	0	0	0	0	X	X	1	X	1
1	0	0	0	0	0	X	1	0	X	0	X
1	0	1	1	1	0	X	0	1	X	X	1
1	1	0	0	0	0	X	1	X	1	0	X
1	1	1	0	0	0	X	1	X	1	X	1



Answer 2



PRESENT STATE			NEXT STATE			FF2		FF1		FF0	
C	B	A	C	B	A	Jc	Kc	Jb	Kb	Ja	Ka
0	0	0	0	1	0	0	X	1	X	0	X
0	0	1	X	X	X	X	X	X	X	X	X
0	1	0	1	0	1	1	X	X	1	1	X
0	1	1	X	X	X	X	X	X	X	X	X
1	0	0	X	X	X	X	X	X	X	X	X
1	0	1	1	1	0	X	0	1	X	X	1
1	1	0	0	0	0	X	1	X	1	0	X
1	1	1	X	X	X	X	X	X	X	X	X

	$\bar{A}$	A
$\bar{B} \bar{C}$		x
$\bar{B} C$	x	x
$B C$		x
$B \bar{C}$	1	x

$J_a = B\bar{C}$

	$\bar{A}$	A
$\bar{B} \bar{C}$	x	x
$\bar{B} C$	x	1
$B C$	x	x
$B \bar{C}$	x	x

$K_a = 1$

	$\bar{A}$	A
$\bar{B} \bar{C}$	1	x
$\bar{B} C$	x	1
$B C$	x	x
$B \bar{C}$	x	x

$J_b = 1$

	$\bar{A}$	A
$\bar{B} \bar{C}$	x	x
$\bar{B} C$	x	x
$B C$	1	x
$B \bar{C}$	1	x

$K_b = 1$

	$\bar{A}$	A
$\bar{B} \bar{C}$		x
$\bar{B} C$	x	x
$B C$	x	x
$B \bar{C}$	1	x

$J_c = B$

	$\bar{A}$	A
$\bar{B} \bar{C}$	x	x
$\bar{B} C$	x	
$B C$	1	x
$B \bar{C}$	x	x

$K_c = B$

Summary

- ✎ These steps can be applied to any sequential circuit to create a counter
 - Specify the counter sequence and draw the state diagram
 - Derive the next-state table from the state diagram
 - Using the J-K FF transition/excitation table, develop the transition table showing the FF inputs required for each transition
 - Transfer J and K states to individual K-maps – there is a K-map for each input of each FF
 - Group K-map cells to simplify and derive the logic expression for each FF input
 - Implement the expressions with combinational logic and combine with FF to create the counter